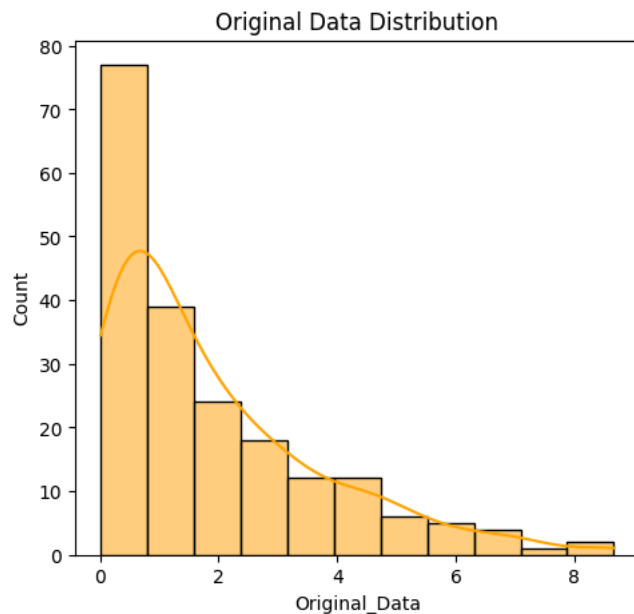


```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

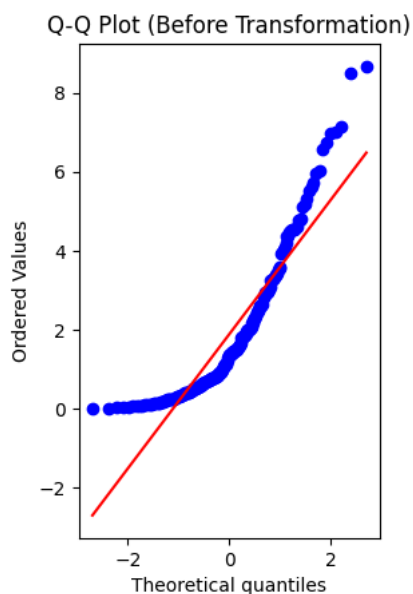
```
np.random.seed(42)
data = np.random.exponential(scale=2, size=200)
df = pd.DataFrame({'Original_Data': data})
```

```
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.histplot(df['Original_Data'], kde=True, color='orange')
plt.title("Original Data Distribution")
```

Text(0.5, 1.0, 'Original Data Distribution')



```
plt.subplot(1,2,2)
stats.probplot(df['Original_Data'], dist="norm", plot=plt)
plt.title("Q-Q Plot (Before Transformation)")
plt.show()
```



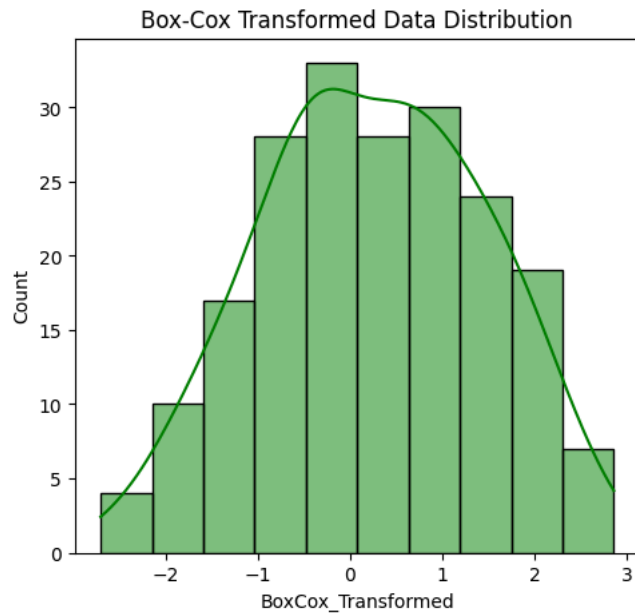
```
transformed_data, best_lambda = stats.boxcox(df['Original_Data'])
df['BoxCox_Transformed'] = transformed_data
```

```
print(f"Optimal Lambda Value: {best_lambda:.4f}")
```

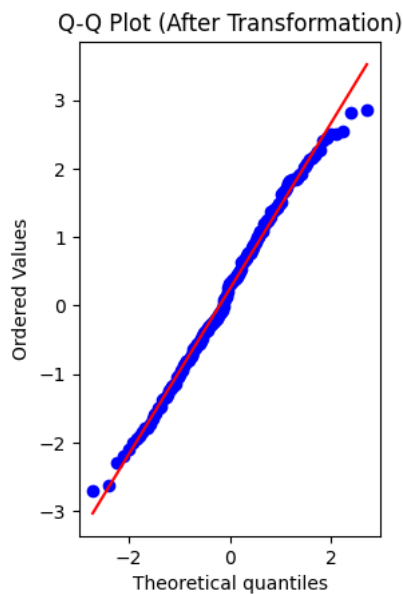
Optimal Lambda Value: 0.2495

```
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.histplot(df['BoxCox_Transformed'], kde=True, color='green')
plt.title("Box-Cox Transformed Data Distribution")
```

Text(0.5, 1.0, 'Box-Cox Transformed Data Distribution')



```
plt.subplot(1,2,2)
stats.probplot(df['BoxCox_Transformed'], dist="norm", plot=plt)
plt.title("Q-Q Plot (After Transformation)")
plt.show()
```



```
stat_original, p_original = stats.shapiro(df['Original_Data'])
stat_transformed, p_transformed = stats.shapiro(df['BoxCox_Transformed'])
```

```
print("Before Transformation:")
print(f"Shapiro-Wilk Statistic = {stat_original:.4f}, p-value = {p_original:.4f}")
```

Before Transformation:
Shapiro-Wilk Statistic = 0.8522, p-value = 0.0000

```
print("\nAfter Box-Cox Transformation:")
print(f"Shapiro-Wilk Statistic = {stat_transformed:.4f}, p-value = {p_transformed:.4f}")
```

After Box-Cox Transformation:
Shapiro-Wilk Statistic = 0.9904, p-value = 0.2075